

Aprendizagem bayesiana aplicada à vigilância epidemiologia das doenças transmissíveis

Leo Bastos



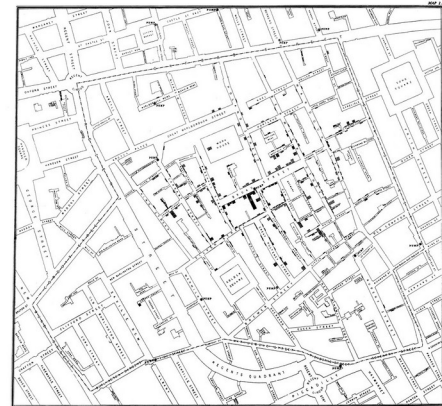
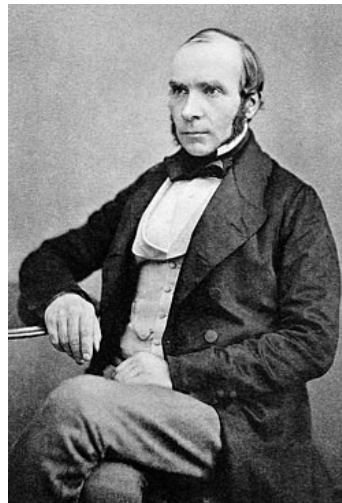
Summary

- **Bayesian epidemiology**
- **An introduction to dengue**
- **Very short term forecast (Predicting now)**
 - Delay-correction models (nowcasting)
- **One-year ahead forecasts**
 - Baseline model to build probabilistic epidemic thresholds
 - Opportunities and challenges for dengue forecasting models

When Bayes meets John Snow



- **Thomas Bayes (1701-1761)**
- $p(\theta \mid x) \propto p(\theta)L(\theta; x)$
- **Father of Bayesianism**



- **John Snow (1813-1856)**
- **Studied cholera in London**
- **Father of epidemiology**

When Bayes meets John Snow



Generalised linear mixed models

Likelihood $Y_t \mid \mu_t, \phi \sim F(\mu_t, \phi), \quad t = 1, 2, 3, \dots, T$

$$g(\mu_t) = \alpha + x_t^T \beta + \gamma_{R[t]}.$$

Prior $\alpha \sim N(0, V_\alpha) \quad \beta \sim N(0, V_\beta) \quad \{\gamma_r\} \sim N(0, Q_\gamma)$

Posterior $p(\alpha, \beta, \{\gamma_r\}, \phi \mid y_1, y_2, \dots, y_T)$



The inference/learning process with respect to the parameters is based on the posterior.

Timeline of Bayesian computation

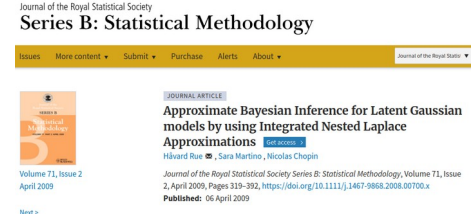
The Journal of Chemical Physics



Metropolis et al. (1953)



Gelfand & Smith (1991)



Rue, Martino & Chopin (2009)

Hastings (1970)

WinBUGS (1997)



JAGS (2007)



STAN (2012)



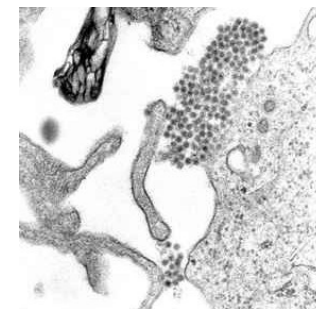
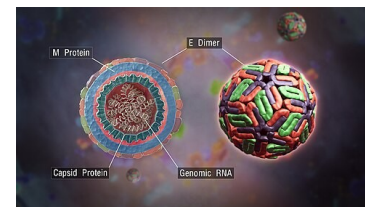
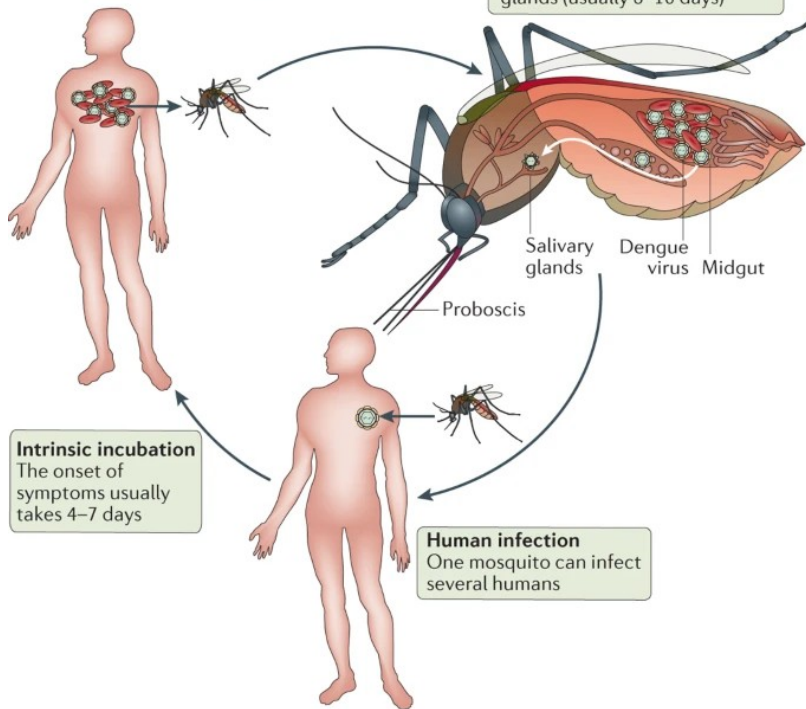
Dengue cycle

Symptoms



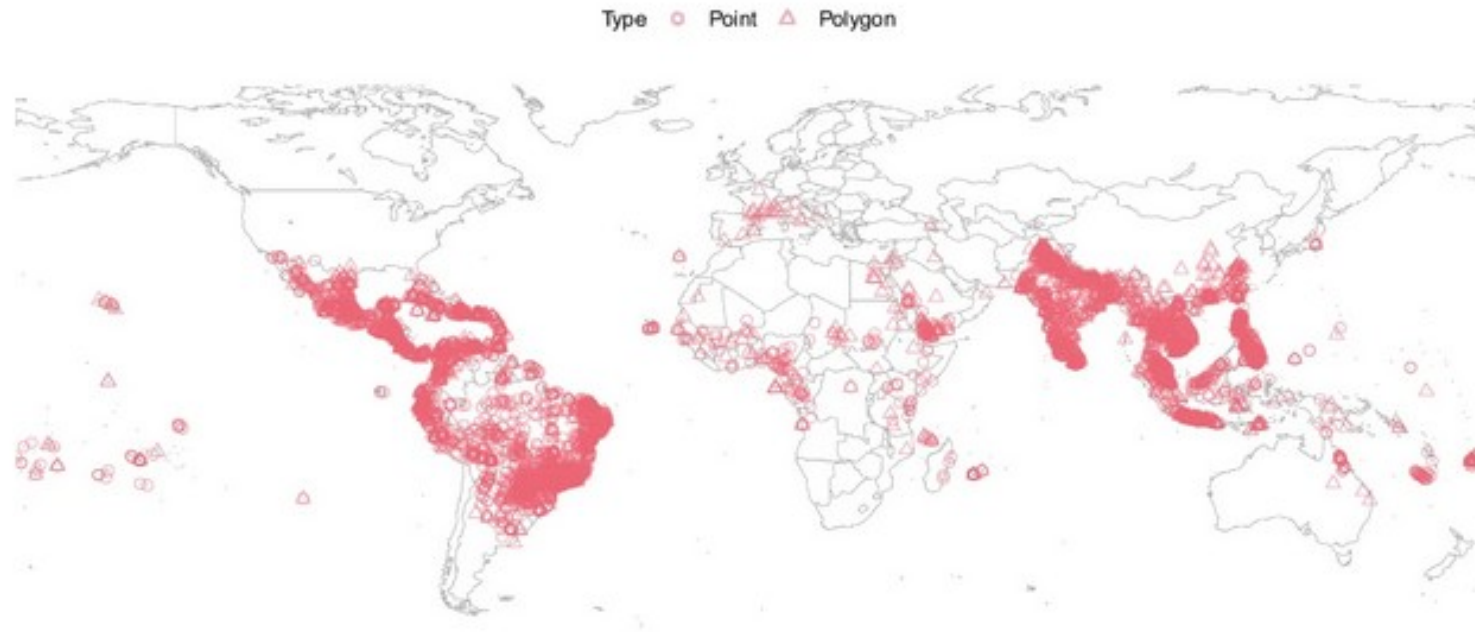
Mosquito infection
Mosquito takes a blood meal from a person with acute dengue

Extrinsic incubation
Virus infects the midgut and eventually travels to the salivary glands (usually 8–10 days)



Dengue occurrence in the world

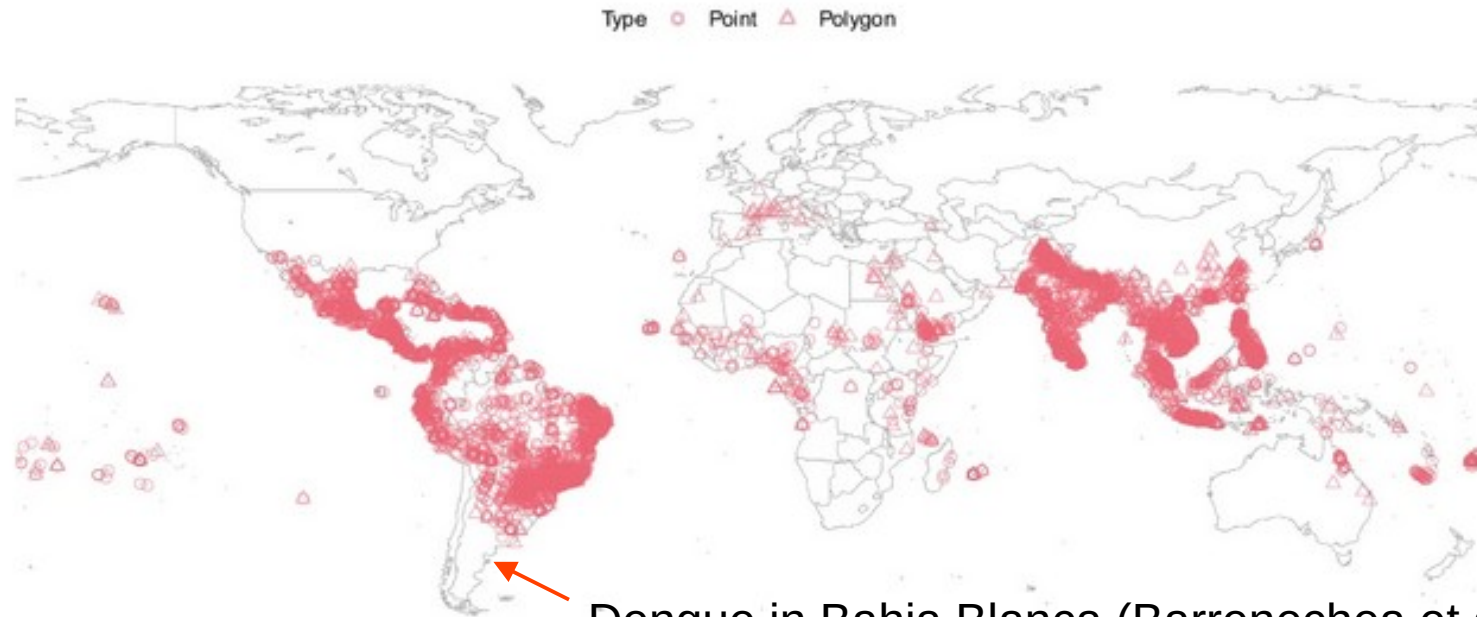
b Dengue



Lim et al. (2025, Nature communications)

Dengue occurrence in the world

b Dengue



Dengue in Bahia Blanca (Barrenechea et al. 2025)

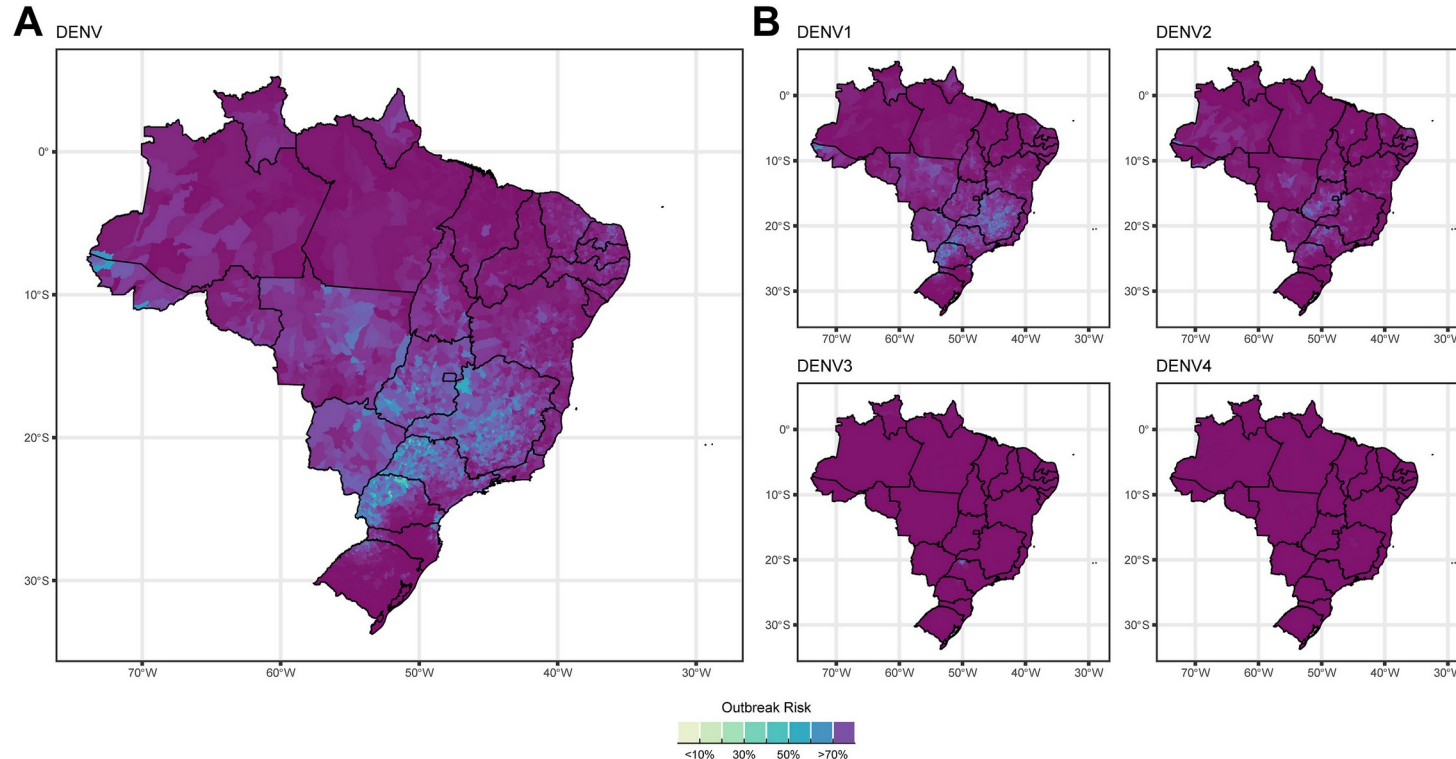
Lim et al. (2025, Nature communications)

Dengue immunity

- **There are four serotypes (DENV1-DENV4)**
 - long-term homotypic immunity
 - short-term heterotypic immunity
- **Vaccines**
 - Dengvaxia (Sanofi) {Recommended in the US, 3 doses; 9-16y*}
 - Qdenga (Takeda) {Recommended by the WHO; 2 doses; 6-59y}
 - Butantan-DV (Butantan) {Approved by ANVISA; 1 dosis; 2-59y}
- **ADE (antibody-dependent enhancement)**



Dengue outbreak risk based on possible susceptibles



THE LANCET Regional Health
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COMMENT • Volume 45, 101068, May 2025

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On the verge: outbreak risk after two years of record-breaking dengue epidemics in Brazil

Rafael Lopes ^a • Leonardo S. Bastos ^b

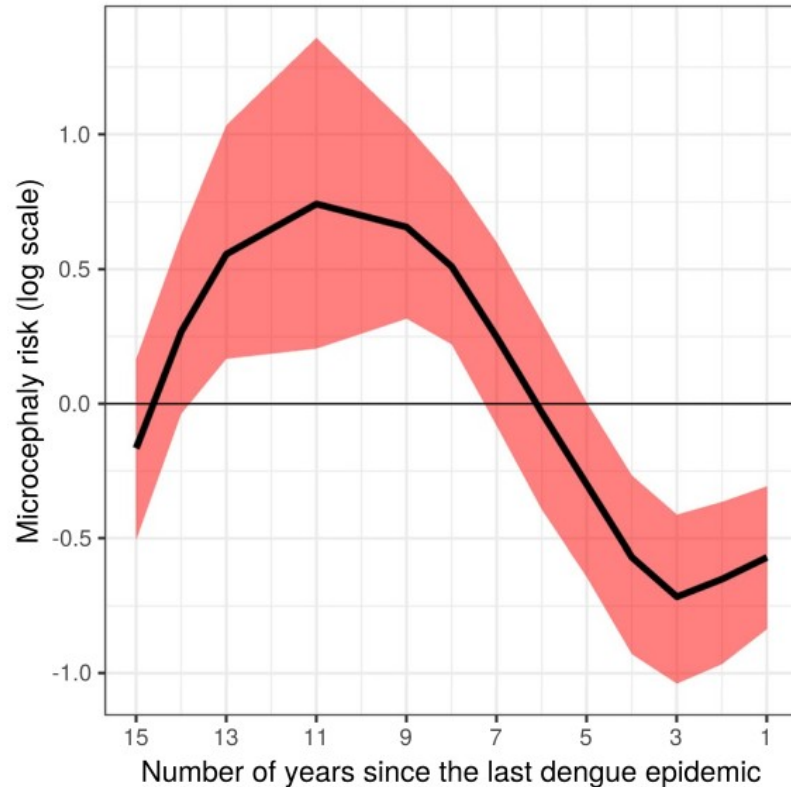
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Lopes & Bastos (2025)



Population immunity: Dengue and CZS

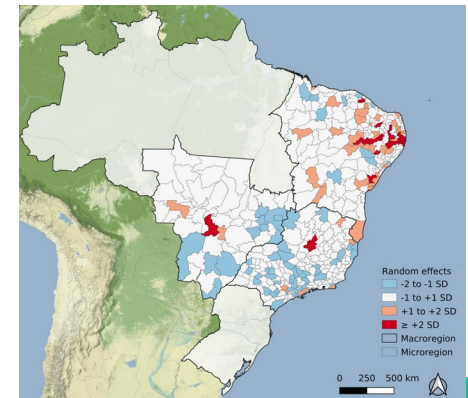
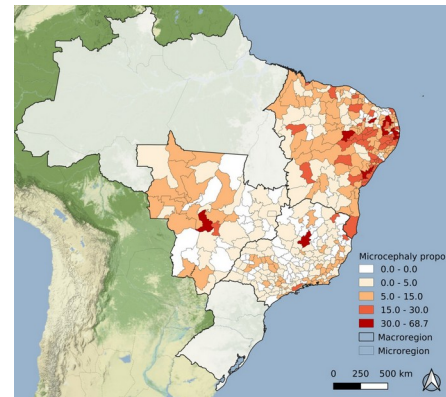


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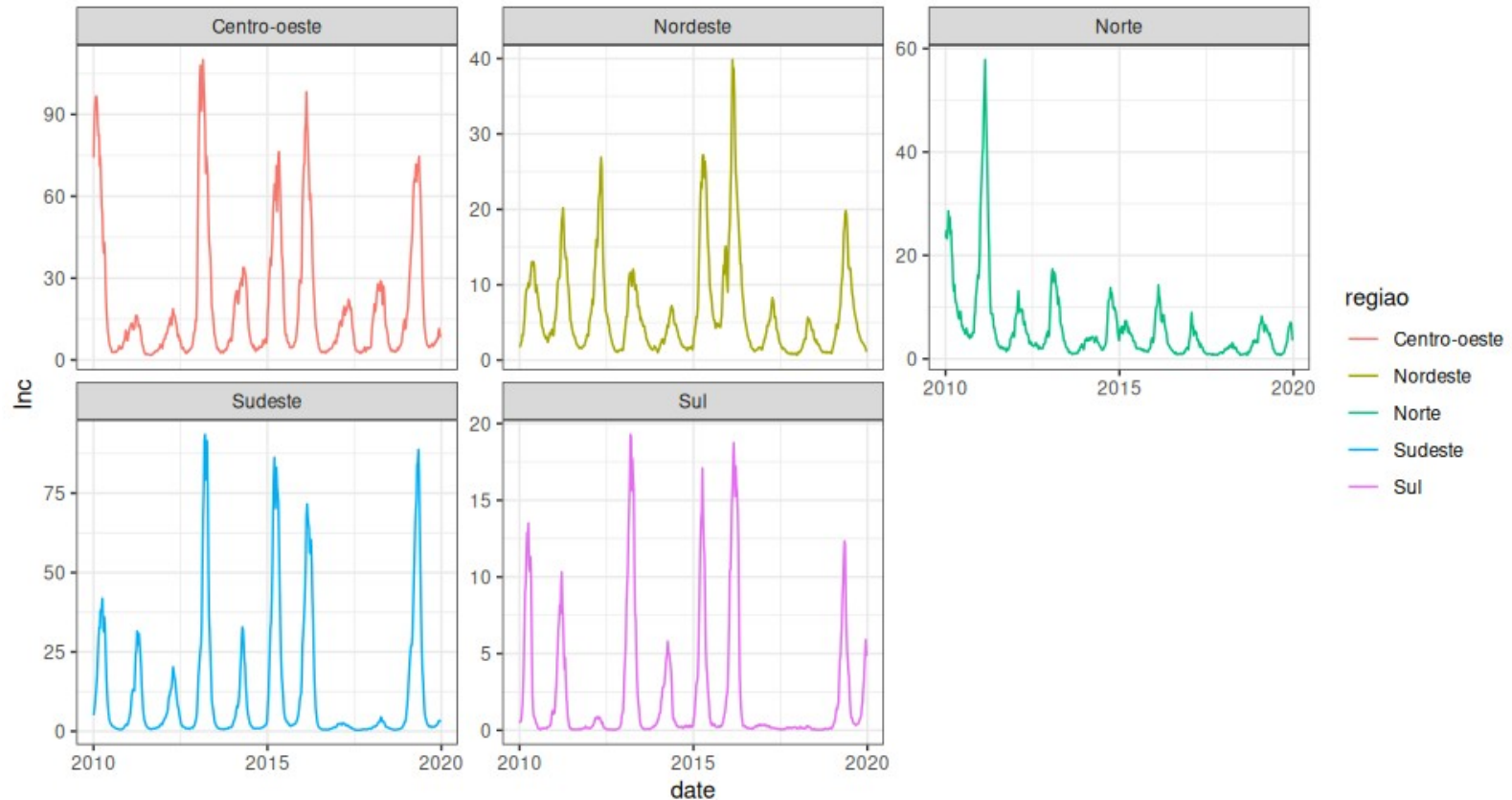
Association of past dengue fever epidemics with the risk of Zika microcephaly at the population level in Brazil

Marília Sá Carvalho^{1*}, Laís Picinini Freitas², Oswaldo Gonçalves Cruz¹, Patrícia Brasil³ & Leonardo Soares Bastos¹

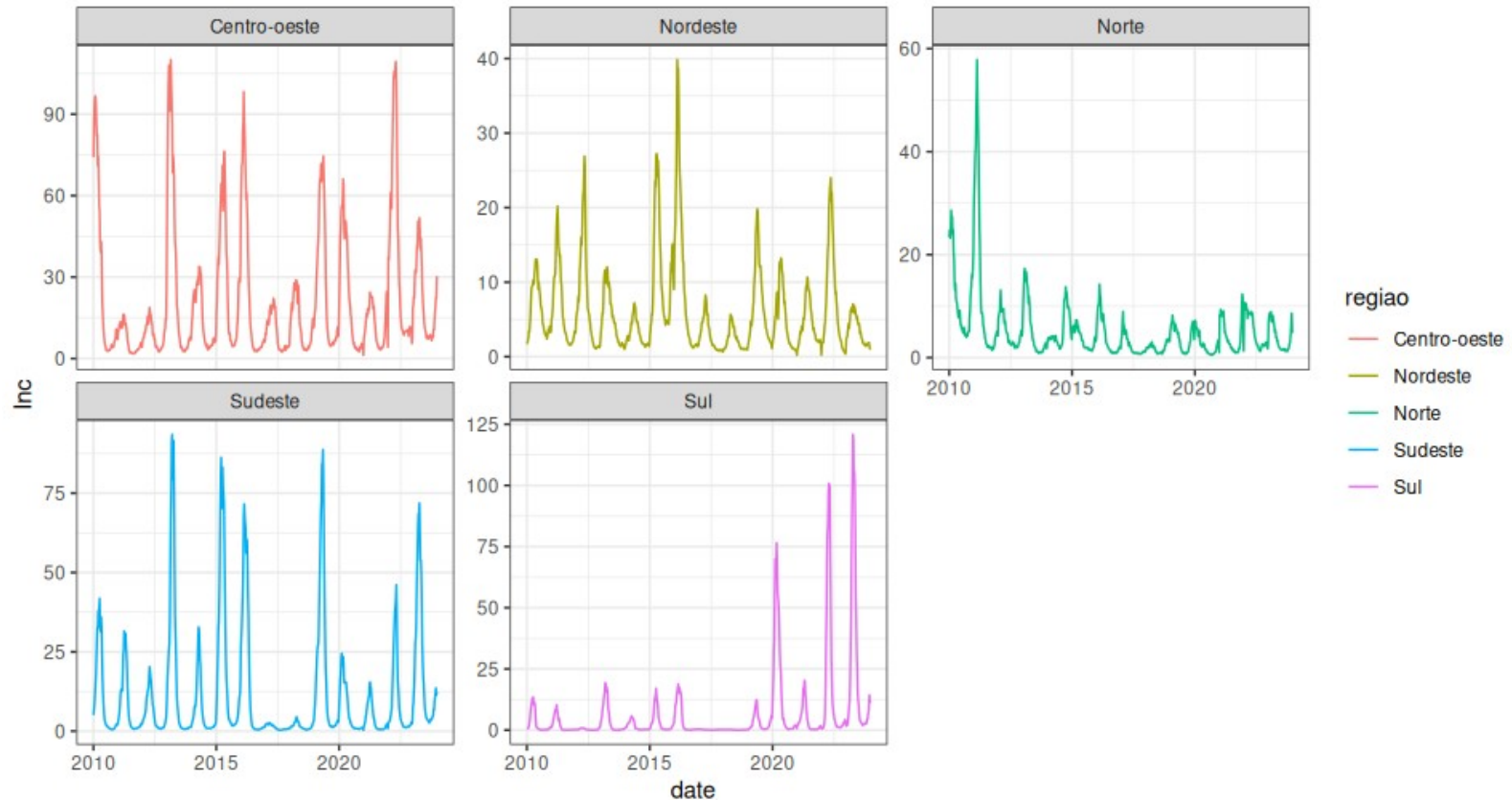
Carvalho et al. (2020)



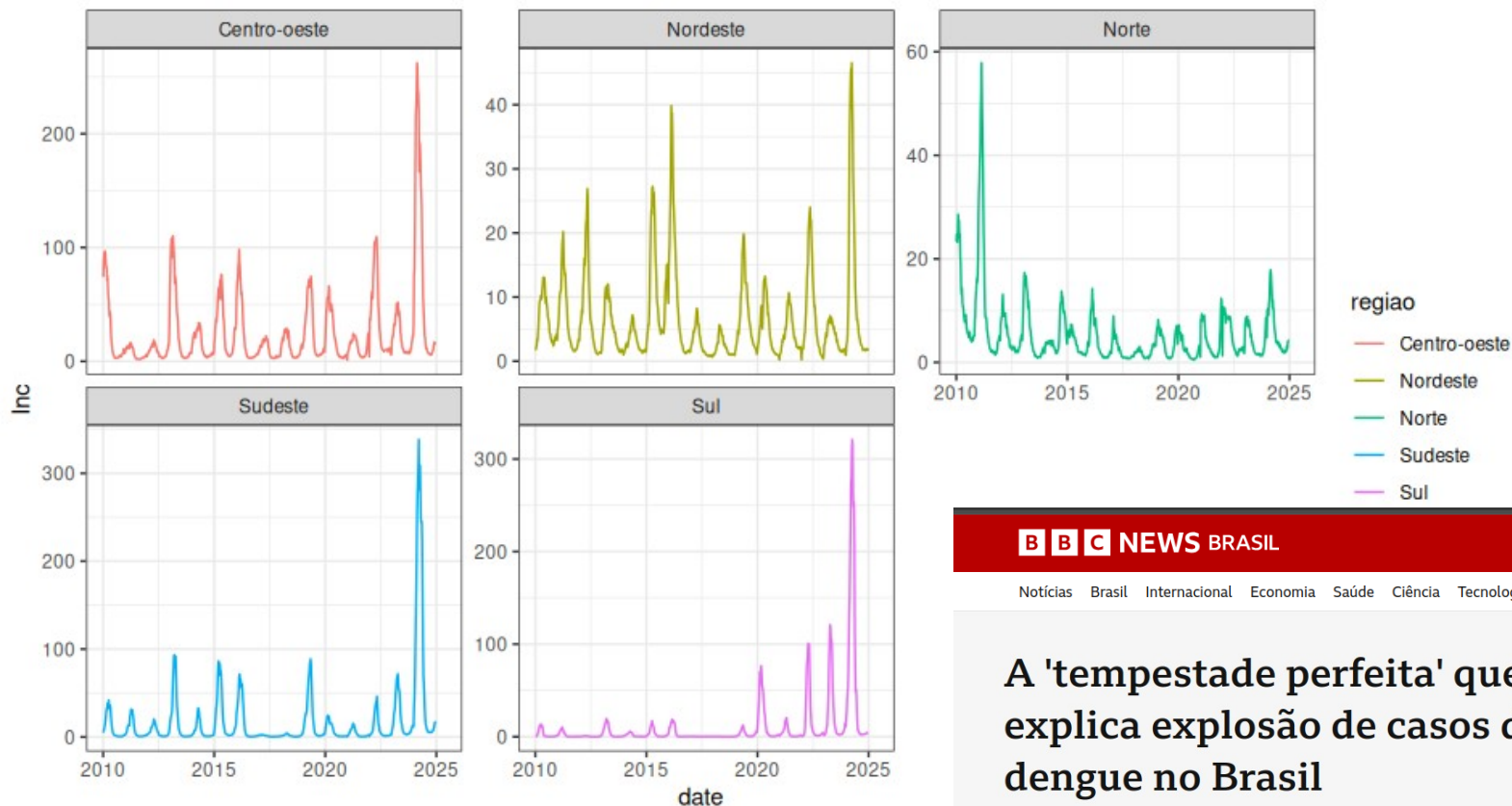
Dengue in Brazil (2010-2020)



Dengue in Brazil (2010-2023)



Dengue in Brazil (2010-2024)



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A 'tempestade perfeita' que explica explosão de casos de dengue no Brasil

Principa

Ibama at
pesquisa
Amazon
agora

Dengue expansion



THE LANCET *Regional Health*
Americas



COMMENT • Volume 12, 100274, August 2022

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Fast expansion of dengue in Brazil

Claudia T. Codeço ^a   • Sara S. Oliveira ^a • Danielle A.C. Ferreira ^b • Thais I.S. Riback ^a • Leonardo S. Bastos ^a • Raquel M. Lana ^{a,c} • et al. [Show more](#)

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Codeço et al. (2022)

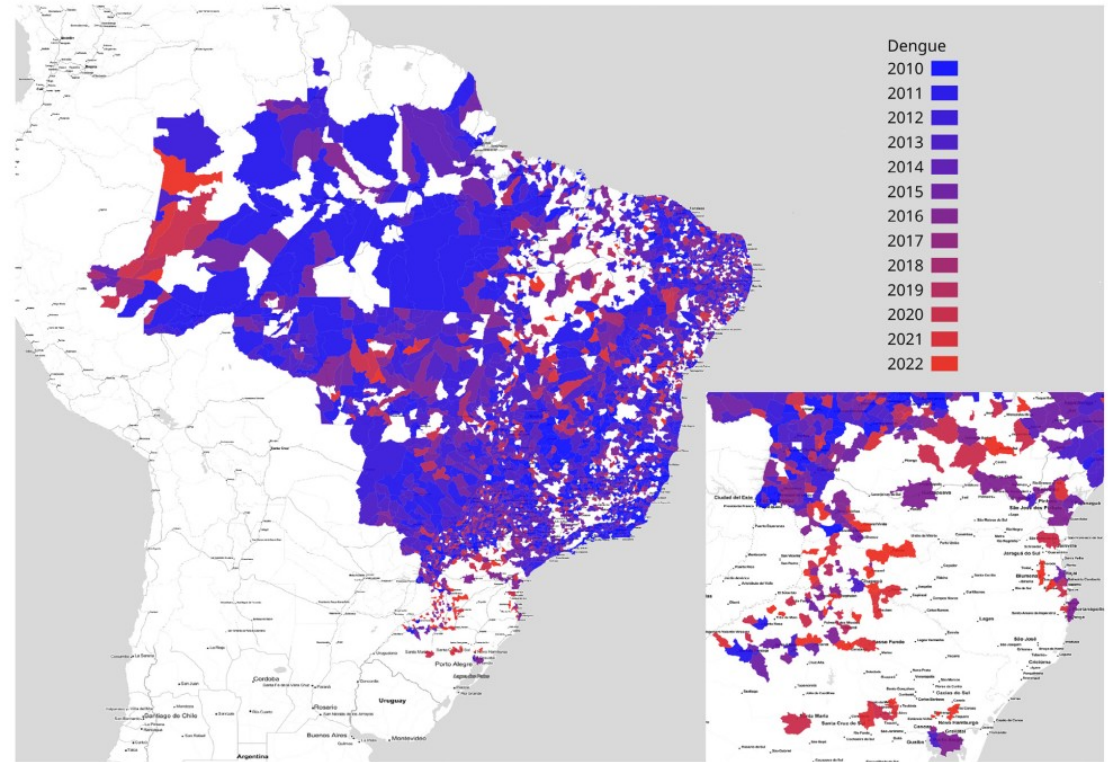


Figure 1. Map of Brazil showing the spread of dengue transmission into previously disease free regions (pre-2010 to March 2022). Source: Infodengue.

Dengue in Bahia Blanca, Argentina



Guillermo Barrenechea

THE LANCET *Regional Health Americas*

ARTICLES • Volume 51, 101254, November 2025

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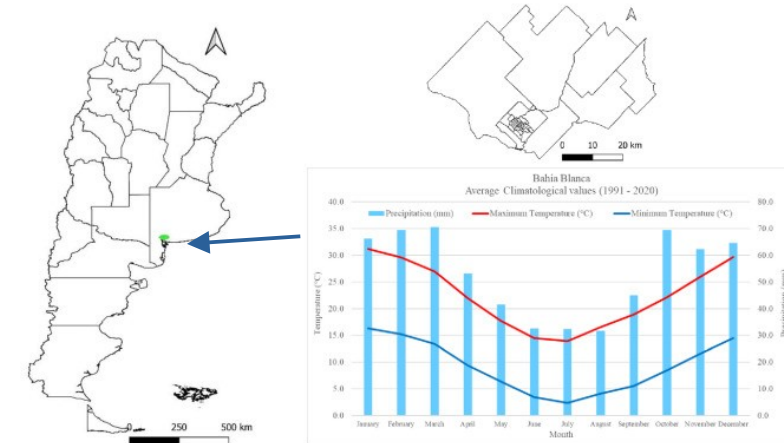
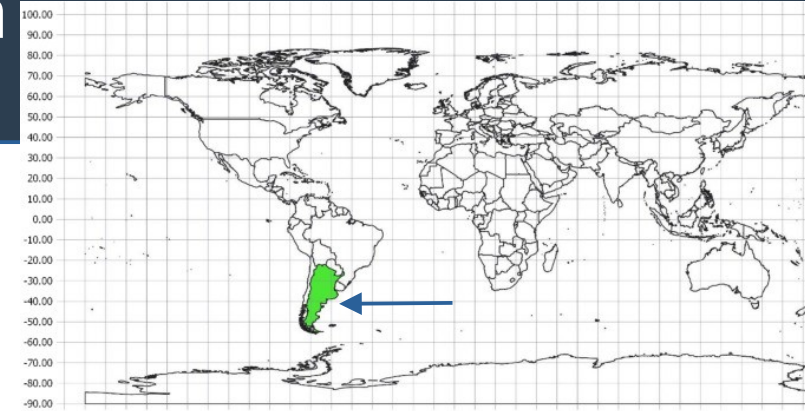
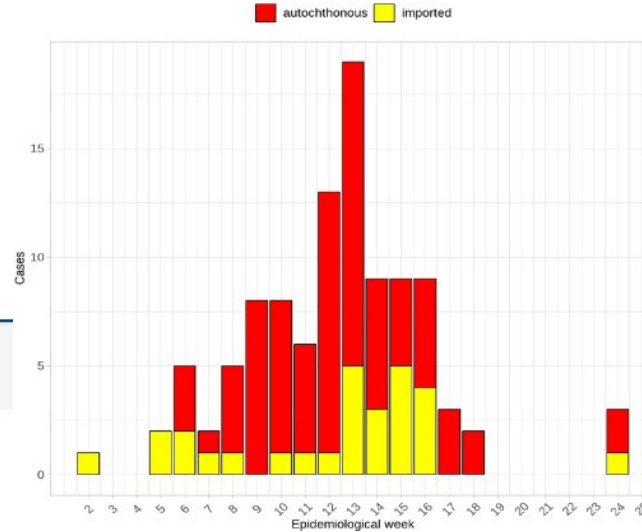
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Characterization of the most austral autochthonous dengue outbreak reported in the world (city of Bahía Blanca, Argentina, January–June 2024). A cross-sectional study

Guillermo Gabriel Barrenechea ^{a,b} • Rocío Sánchez ^c • Ornella Cynthia Calderón ^d • Ignacio Rodrigo Buffone ^d • Leonardo Soares Bastos ^{e,f}

Affiliations & Notes ▾ Article Info ▾

Barrenechea et al. (2025)

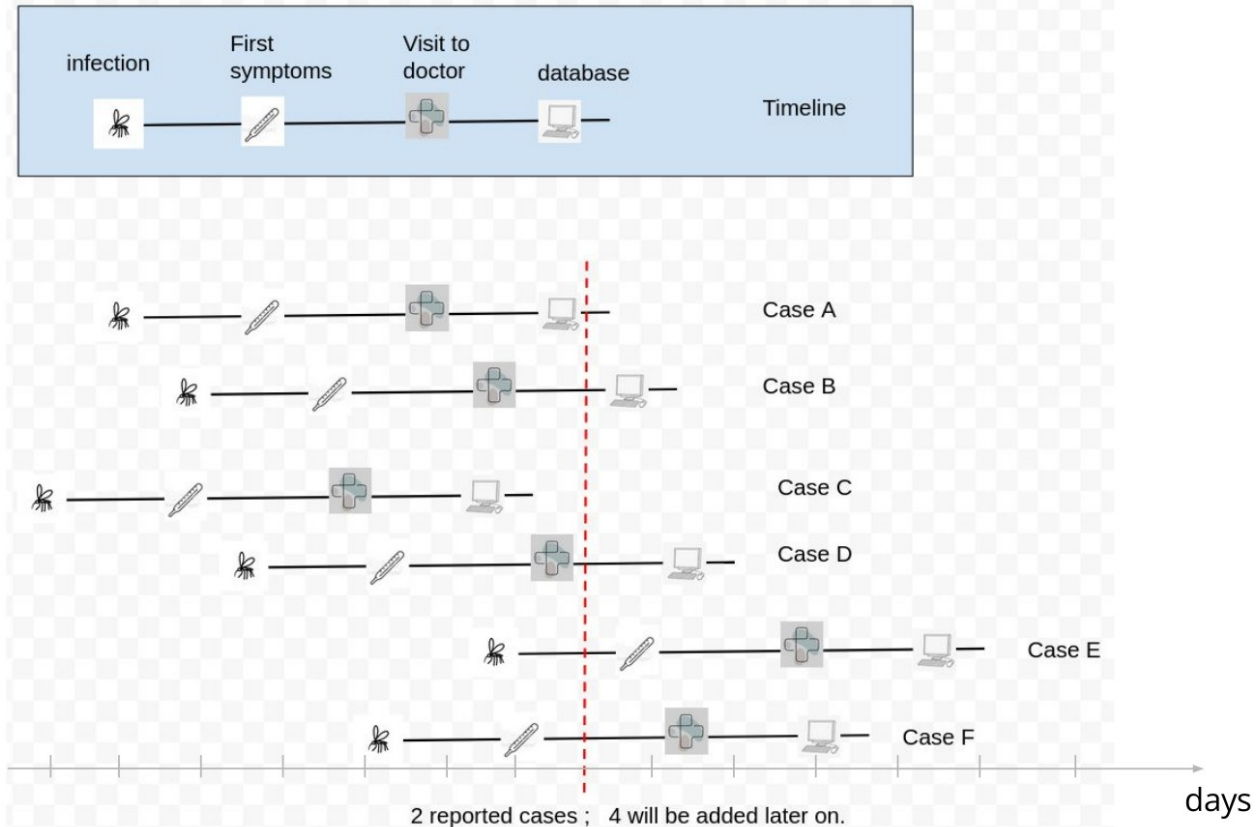


Case definitions (in Brazil)

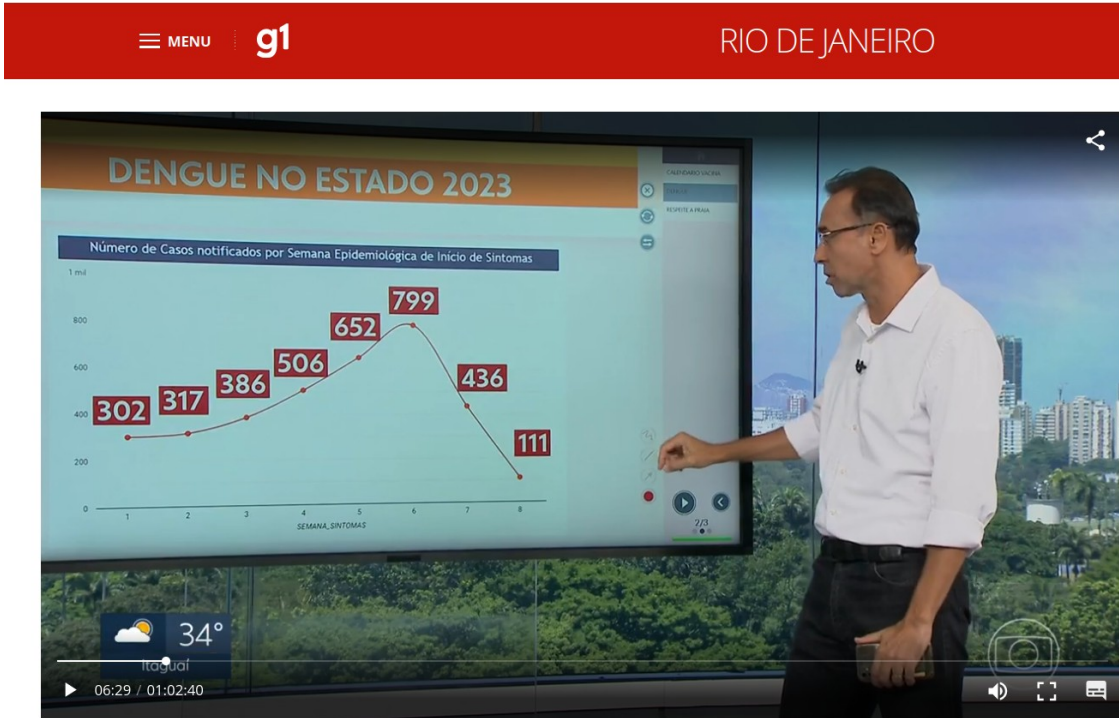
- Any **suspected case** must be reported
 - It is possible a non dengue case being reported as dengue!
 - e.g. chikungunya, zika, oropouche,...
- A notified case is not discarded is called **probable case**.
- If the case is tested, it is then called **confirmed case**.
- Serotype or genotyped could be know from a confirmed case
 - A tiny proportion of the suspected cases would be sequenced.

The notification process

Modeling the observation delay process



Notification delay problem



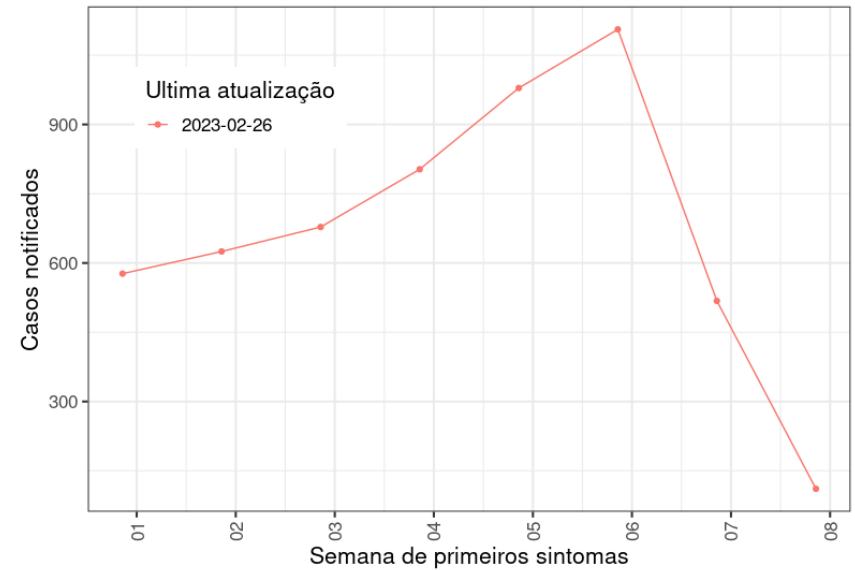
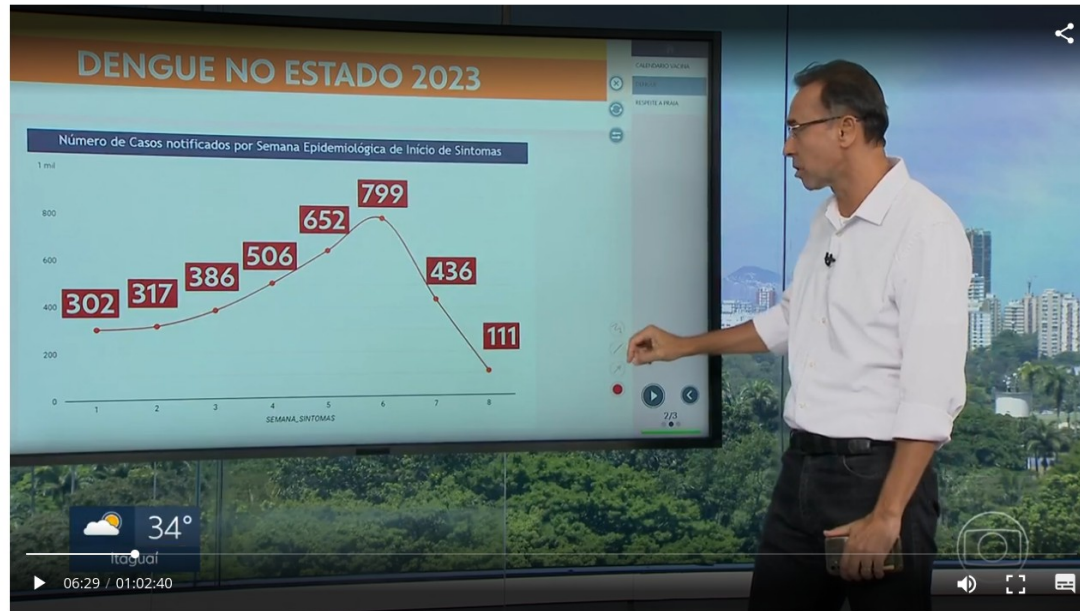
- **Dengue in Rio de Janeiro, 2023**
- End of week 9 (First week of March)
- The TV presenter emphasizes
 - that the **peak occurred** on week 6
 - the “good news” cases are **dropping**

Rio, 2023-03-04

Notification delay problem

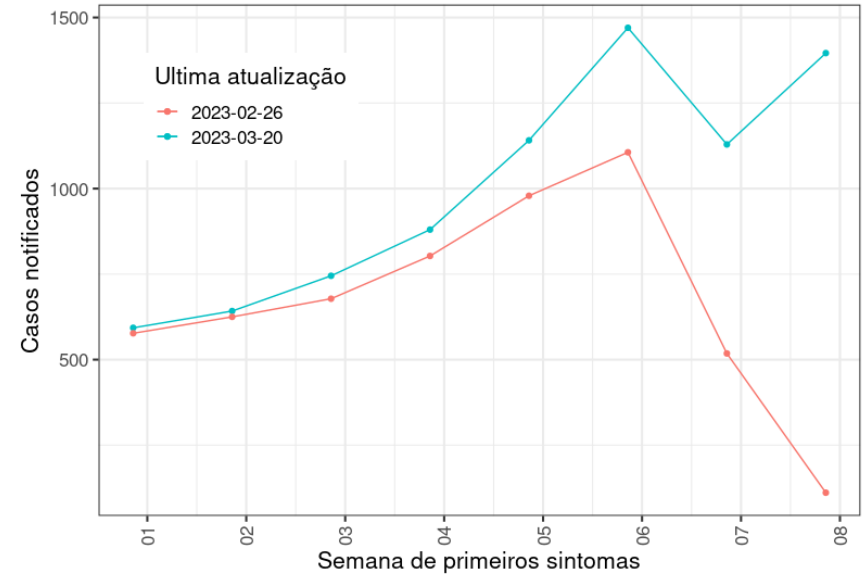
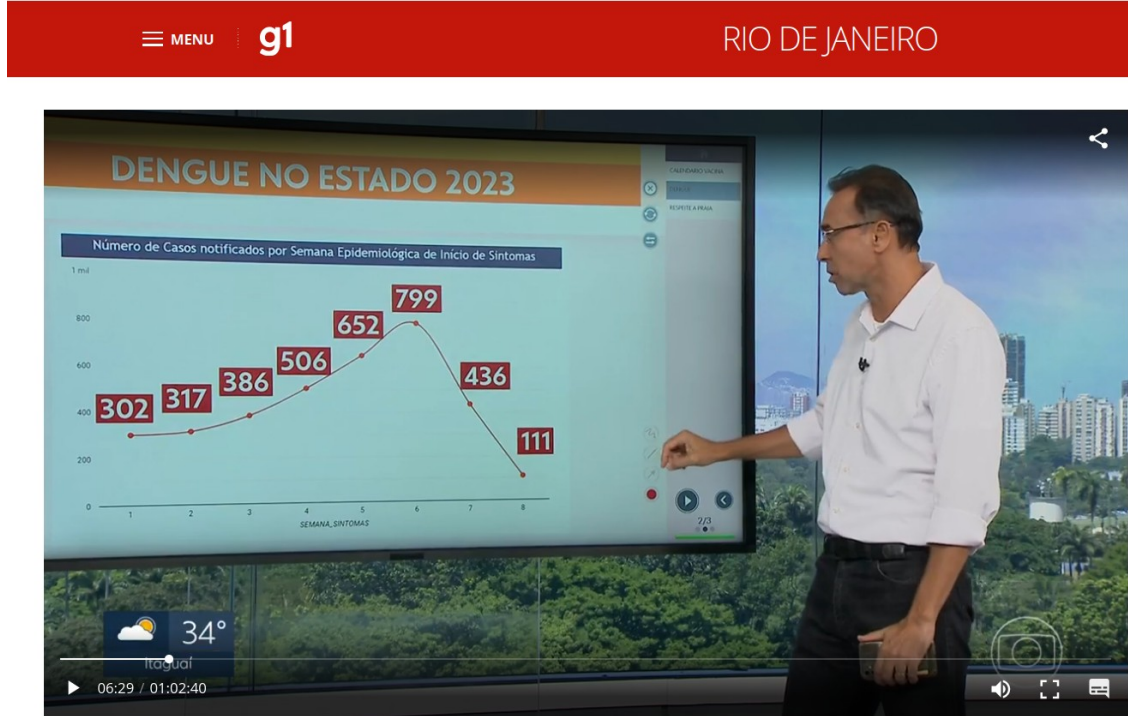
MENU g1

RIO DE JANEIRO



Rio, 2023-03-04

Notification delay problem



Rio, 2023-03-04

Bayesian chain-ladder model

$$n_{t,d} \sim \text{NegBin}(\lambda_{t,d}, \phi)$$

- Bastos et al. (2019) (INLA)

$$\log(\lambda_{t,d}) = \alpha + \beta_t + \gamma_d$$

- McGough et al. (2020) (MCMC)

$$\lambda_{t,d} = \delta_d e^{\beta_t}$$

$\{n_{t,d}\} = \{n_{t,d} : T \leq t + d\}$

Time	0	1	2	...	D-2	D-1	D	N
1	$n_{1,0}$	$n_{1,1}$	$n_{1,2}$		$n_{1,D-2}$	$n_{1,D-1}$	$n_{1,D}$	N_1
2	$n_{2,0}$	$n_{2,1}$	$n_{2,2}$		$n_{2,D-2}$	$n_{2,D-1}$	$n_{2,D}$	N_2
3	$n_{3,0}$	$n_{3,1}$	$n_{3,2}$		$n_{3,D-2}$	$n_{3,D-1}$	$n_{3,D}$	N_3
...								
T-D	$n_{T-D,0}$	$n_{T-D,1}$	$n_{T-D,2}$		$n_{T-D,D-2}$	$n_{T-D,D-1}$	$n_{T-D,D}$	N_{T-D}
T-D+1	$n_{T-D+1,0}$	$n_{T-D+1,1}$	$n_{T-D+1,2}$		$n_{T-D+1,D-2}$	$n_{T-D+1,D-1}$	$n_{T-D+1,D}$	N_{T-D+1}
T-D+2	$n_{T-D+2,0}$	$n_{T-D+2,1}$	$n_{T-D+2,2}$		$n_{T-D+2,D-2}$	$n_{T-D+2,D-1}$	$n_{T-D+2,D}$	N_{T-D+2}
T-2	$n_{T-2,0}$	$n_{T-2,1}$	$n_{T-2,2}$		$n_{T-2,D-2}$	$n_{T-2,D-1}$	$n_{T-2,D}$	N_{T-2}
T-1	$n_{T-1,0}$	$n_{T-1,1}$	$n_{T-1,2}$		$n_{T-1,D-2}$	$n_{T-1,D-1}$	$n_{T-1,D}$	N_{T-1}
T	$n_{T,0}$	$n_{T,1}$	$n_{T,2}$	...	$n_{T,D-2}$	$n_{T,D-1}$	$n_{T,D}$	N_T
T+1	$n_{T+1,0}$	$n_{T+1,1}$	$n_{T+1,2}$		$n_{T+1,D-2}$	$n_{T+1,D-1}$	$n_{T+1,D}$	N_{T+1}
T+2	$n_{T+2,0}$	$n_{T+2,1}$	$n_{T+2,2}$		$n_{T+2,D-2}$	$n_{T+2,D-1}$	$n_{T+2,D}$	N_{T+2}
...								
T+K	$n_{T+K,0}$	$n_{T+K,1}$	$n_{T+K,2}$		$n_{T+K,D-2}$	$n_{T+K,D-1}$	$n_{T+K,D}$	N_{T+K}

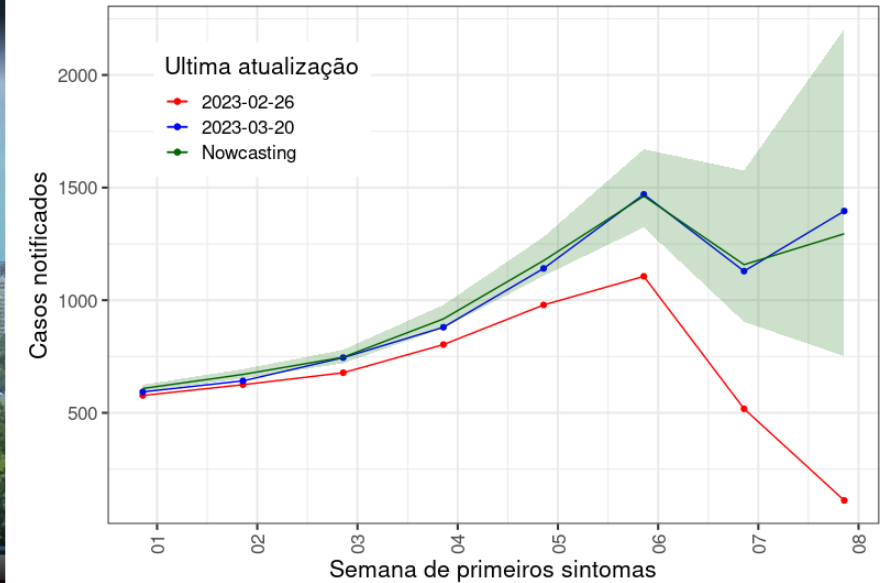
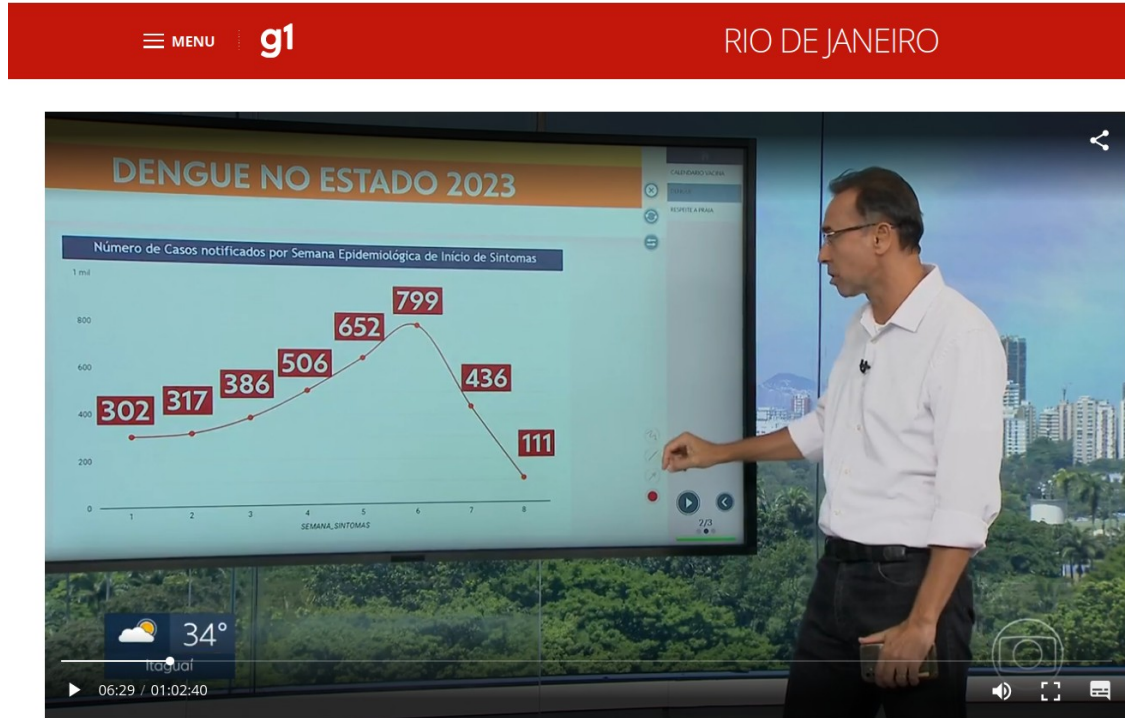
Observations
Nowcasting
Forecasting

$\{n_{t,d}^*\} = \{n_{t,d} : T > t + d\}$

$$P(\{n_{t,d}^*\} \mid \{n_{t,d}\}) = \int_{\theta \in \Theta} p(\theta, \{n_{t,d}^*\} \mid \{n_{t,d}\}) d\theta$$

$$= \int_{\theta \in \Theta} p(\theta \mid \{n_{t,d}\}) p(\{n_{t,d}^*\} \mid \theta) d\theta$$

Notification delay problem



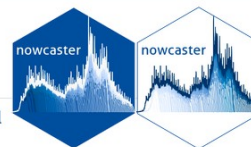
Rio, 2023-03-04

Bayesian Nowcasting

- The Bayesian chain-ladder models is implemented in:
 - two National level disease surveillance systems
 - R package: *nowcaster*



Nowcaster



nowcaster is a R package for “nowcasting” epidemiological time-series. Every single system of notification has an intrinsic delay, *nowcaster* can estimate how many counts of any epidemiological data of interest (*i.e.*, daily cases and deaths counts) by fitting a negative binomial model to the time steps of delay between onset date of the event, (*i.e.*, date of first symptoms for cases or date of occurrence of death) and the date of report (*i.e.*, date of notification of the case or death).

nowcaster is based on the [R-INLA](#) and [INLA](#) packages for “Integrated Nested Laplace Approximation” algorithm to Bayesian inference. [INLA](#) is a fast alternative to others methods for Bayesian inference like **MCMC**. An introduction to [INLA](#) can be found [here](#).

nowcaster is build for epidemiological emergency use, it was constructed for the Brazilian Severe Acute Respiratory Illness (SARI) surveillance database (SIVEP-Gripe).

Installing

Before installing the package certify you have an active installation of [INLA](#), to do so you can run the following code:

```
install.packages("INLA",  
  repos=c(getOption("repos"),  
  INLA="https://inla.r-inla-download.org/R/stable"),  
  dep=TRUE)
```

Links

[Browse source code](#)

[Report a bug](#)

License

[Full license](#)

GPL (>= 3)

Citation

[Citing nowcaster](#)

Developers

[Rafael Lopes](#)

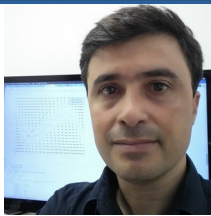
Author, maintainer

[Leonardo Bastos](#)

Author

<https://covid19br.github.io/nowcaster/>

(Nearly) Real-time early warning systems



<https://info.dengue.mat.br/>



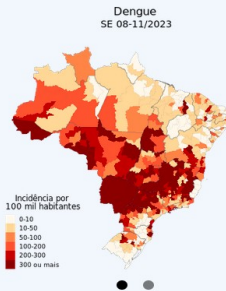
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Estimated cases

Integrated analysis of epidemiological, climate and social media data.

Weekly update:

- Estimated incidence (nowcasting)
- Cities with favorable environment for transmission
- Attention level of municipalities



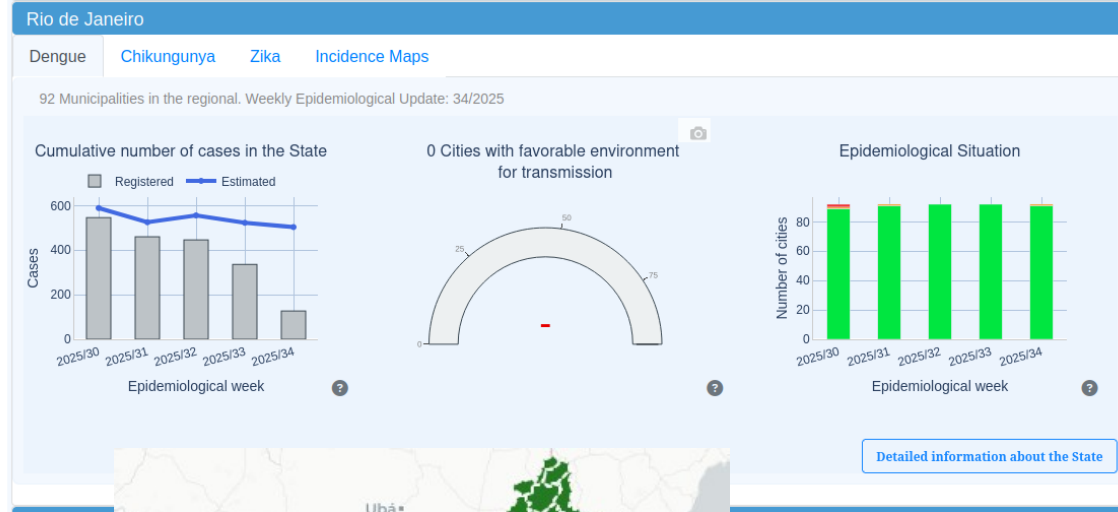
Features:

- Municipal reports
- State map views
- API

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There are many ways to participate
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Weekly updated!



Infodengue



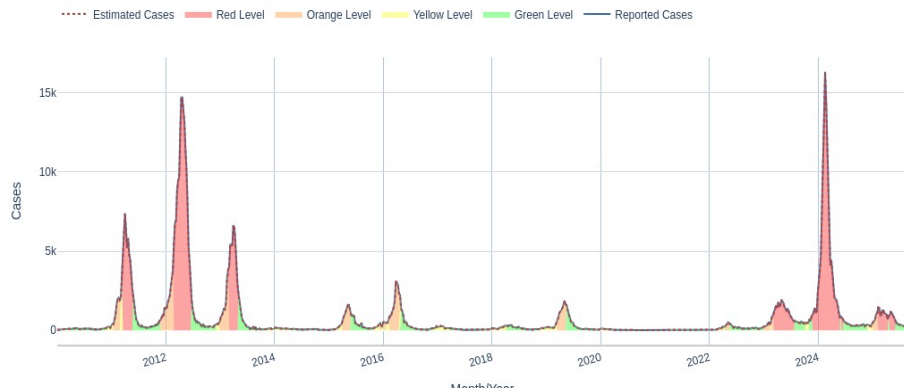
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Dengue Situation - Rio de Janeiro at Aug. 30, 2025

- **Population:** 6.6 million of inhabitants.
- **Estimated Incidence epidemiological week 35:** 3.8 per 100 thousand inhabitants.
- **Arborivirosis Situation report of :** [Rio de Janeiro](#)
- **For more details go to the States page:** [Rio de Janeiro](#)

Select the disease to be shown ☒ Dengue ☐ Chikungunya ☐ Zika

Chance of alert orange or red for Dengue in the municipality of Rio de Janeiro in the next week: 34.0%



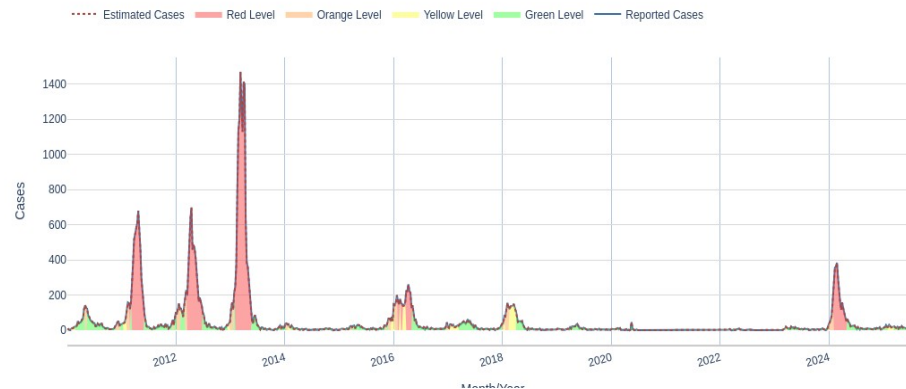
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Dengue Situation - Niterói at Aug. 30, 2025

- **Population:** 523664 of inhabitants.
- **Estimated Incidence epidemiological week 35:** 0.8 per 100 thousand inhabitants.
- **Arborivirosis Situation report of :** [Niterói](#)
- **For more details go to the States page:** [Rio de Janeiro](#)

Select the disease to be shown ☒ Dengue ☐ Chikungunya ☐ Zika

Chance of alert orange or red for Dengue in the municipality of Niterói in the next week: 45.6%



Improving the delay correction

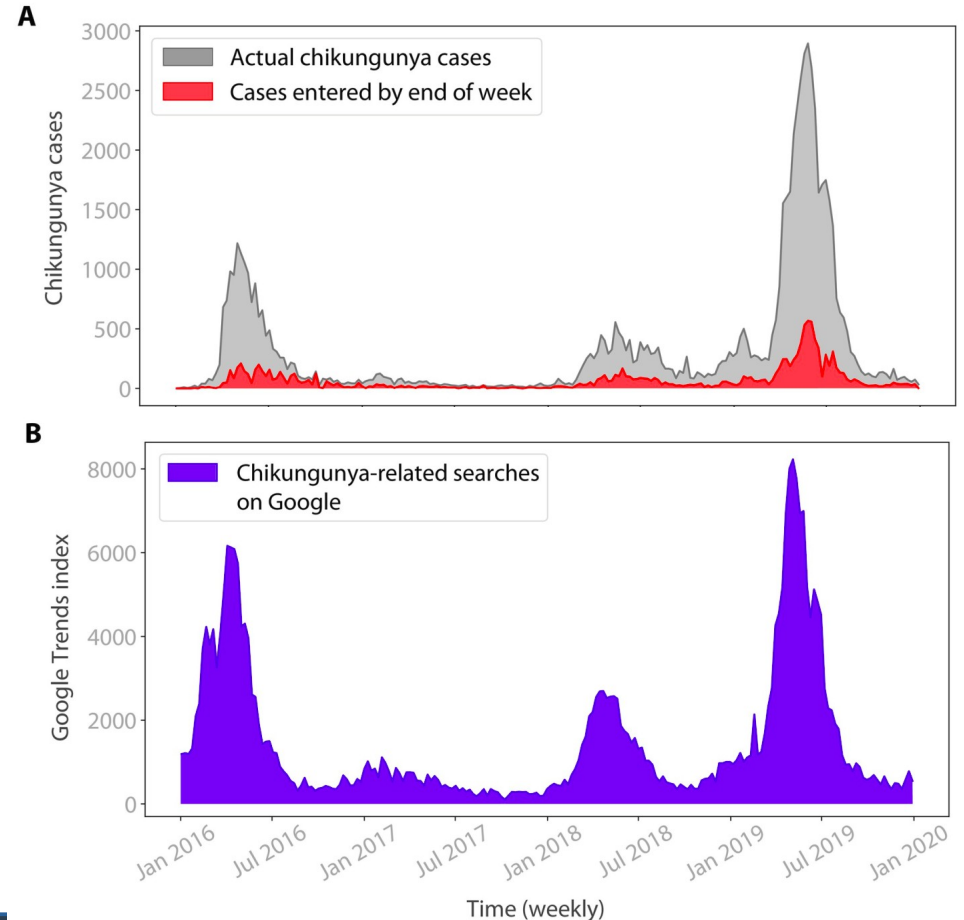
- Covariates may improve corrections

$$n_{t,d} \sim \text{NegBin}(\lambda_{t,d}, \phi)$$

$$\log(\lambda_{t,d}) = \alpha + \beta_t + \gamma_d + \mathbf{x}_{t,d}^T \delta$$

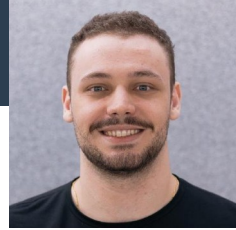
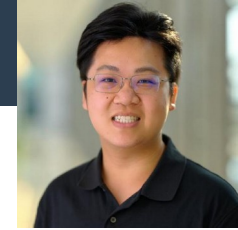
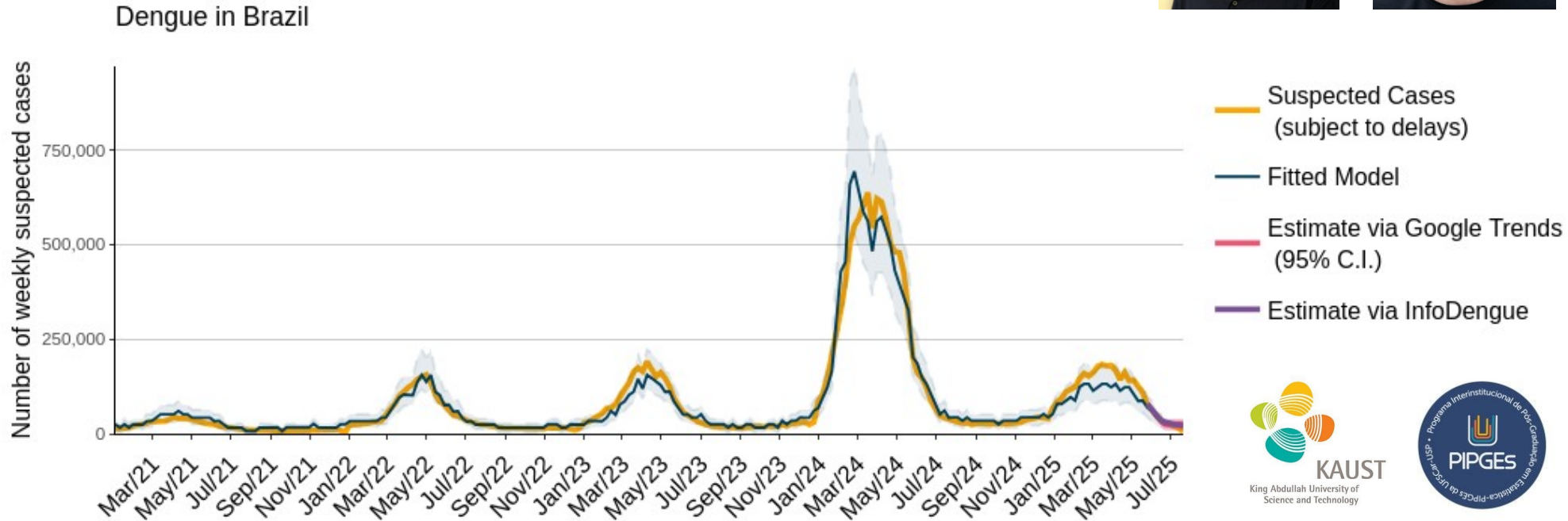
Bastos et al. (2019)

Mizzi (2019, PhD Warwick), Miller et al. (2022)



Nowcasting using google searches

Corrected number of suspected dengue cases via Google Trends

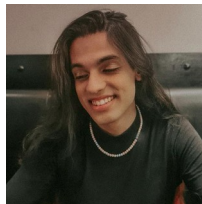


Yang Xian, Guilherme Soares, LB, Rafael Izbick, Paula Moraga (2025, PLOS NTD)

Short-term forecasts

Bayesian additive regression trees (BART)

- *BART*: R package
- Msc dissertation (Jo Arruda)
- 4-week ahead forecasts
- Attribute engineering
- 4 Brazilian cities



Data

- Dengue cases



- climate (ERA5)



- Temperature, humidity

- Google trends



Short-term forecasts for Rio

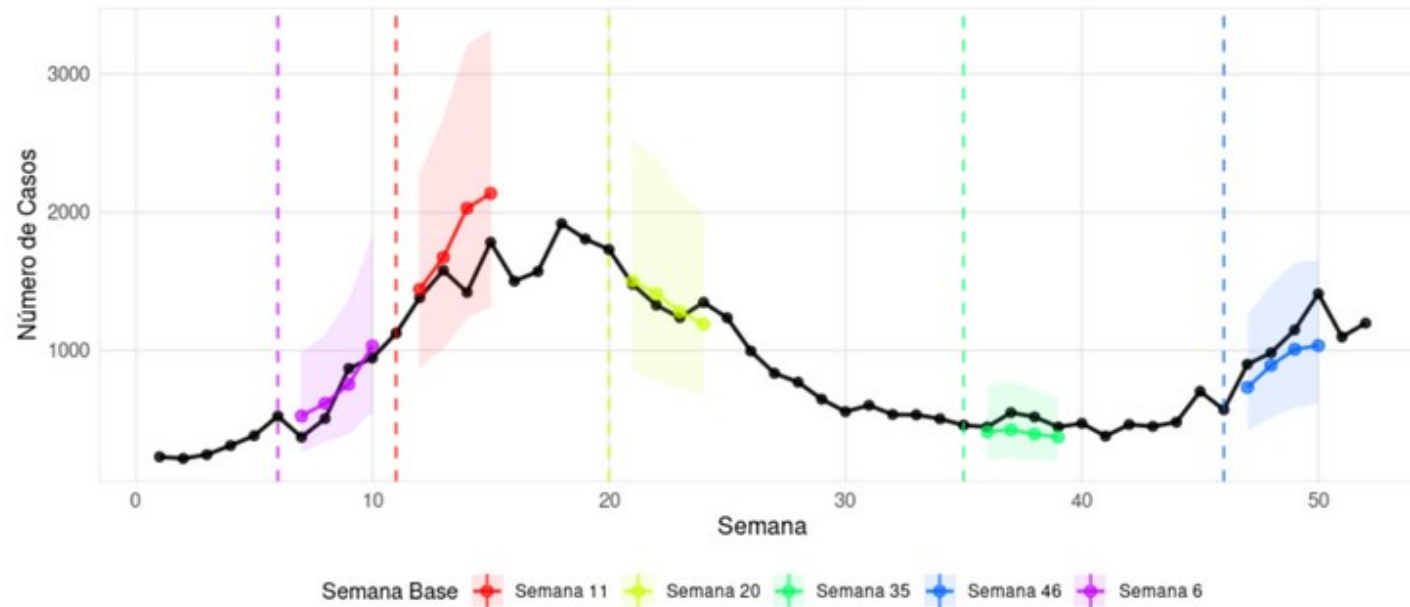


Figura 4.2 – Previsões de casos de dengue originadas de diferentes pontos de partida sobrepostas aos casos reais durante o ano de 2023 - Rio de Janeiro

Arruda (2025)

Multivariate nowcasting

Dengue and Chikungunya joint model

Shared component model

$$n_{t,d}^{(i)} \sim \text{NegBin}(\lambda_{t,d}^{(i)}, \phi^{(i)})$$

$$\log(\lambda_{t,d}^{(i)}) = \alpha + \beta_t + \gamma_d + \delta_t^{(i)} + \psi_d^{(i)}$$

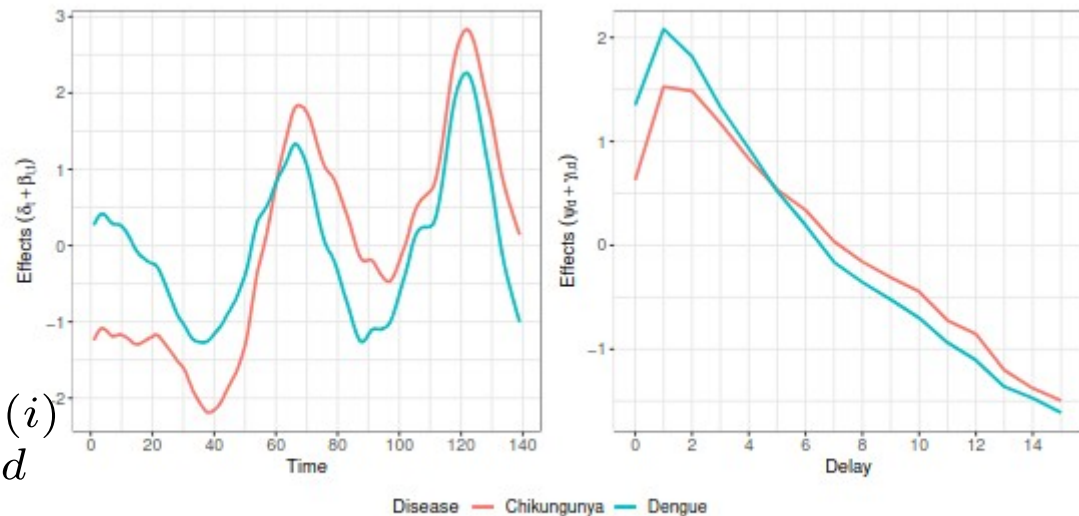
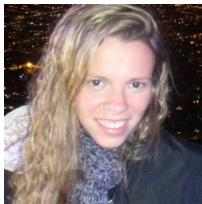
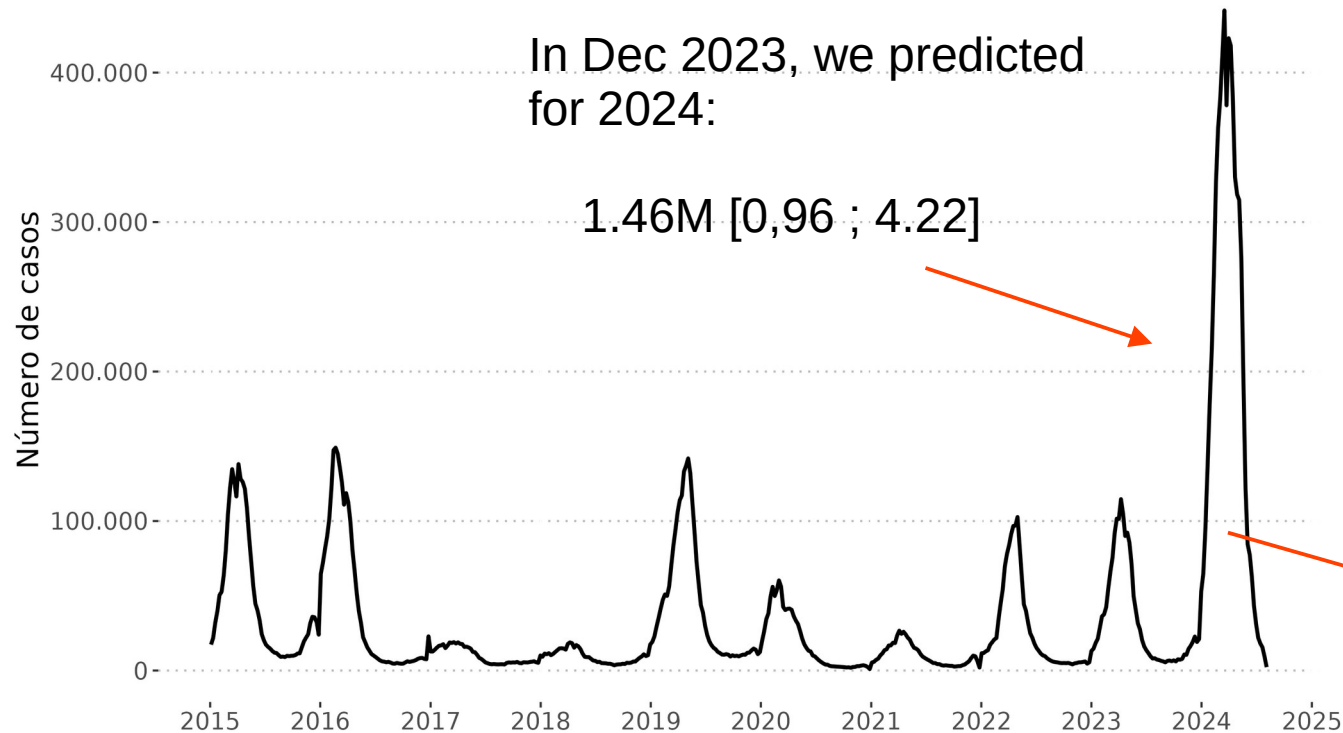


FIGURE 6 Effects of time and delay for dengue (blue) and chikungunya (red).

Dos Santos (2023, Msc, UFRJ)

Dengue in Brazil: Can we predict one-year ahead?



g1

SAÚDE

Brasil pode chegar a 4,2 milhões de casos de dengue em 2024, diz Ministério da Saúde

Os primeiros 521 municípios a receberem a vacina contra a doença devem ser cobertos até o final de março. Crianças de 10 a 11 anos serão primeiro grupo a receber o imunizante.

Por Paloma Rodrigues, TV Globo
09/02/2024 08h34 - Atualizado há um ano

≡

ISTOÉ

Início > Notícias

Brasil pode registrar até 4,2 milhões de casos de dengue em 2024

Agência Brasil — 30/01/24 - 12h24min Em Notícias

> 6 million cases!

Dengue forecasting challenge



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2025 Sprint: 2nd Infodengue-Mosqlimate Dengue Challenge (IMDC)

Welcome to the dengue forecast sprint page

- “1 team in 2023”
- 6 teams in 2024
- 16 teams in 2025

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Sprint 2025 webinar

Webinar – 2nd Infodengue-Mosqlimate Dengue Challenge (IMDC)

Attention, scientific community and public health professionals! On Wednesday, Sep 3, 2025 · 2pm – 3pm (Central European Time), join the virtual meeting of the 2nd Infodengue-Mosqlimate Dengue Challenge (IMDC). We will present the results of the validation round. This webinar will bring together researchers, health professionals, data scientists, and forecasters who have submitted predictive models to address a major global health challenge: predicting the burden of dengue, in the context of current global climate transformations.

[read more](#)

<https://sprint.mosqlimate.org/>

Leveraging probabilistic forecasts for dengue preparedness and control: the 2024 Dengue Forecasting Sprint in Brazil

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This article is a preprint and has not been peer-reviewed [what does this mean?]. It reports new medical research that has yet to be evaluated and so should *not* be used to guide clinical practice.

[Abstract](#)

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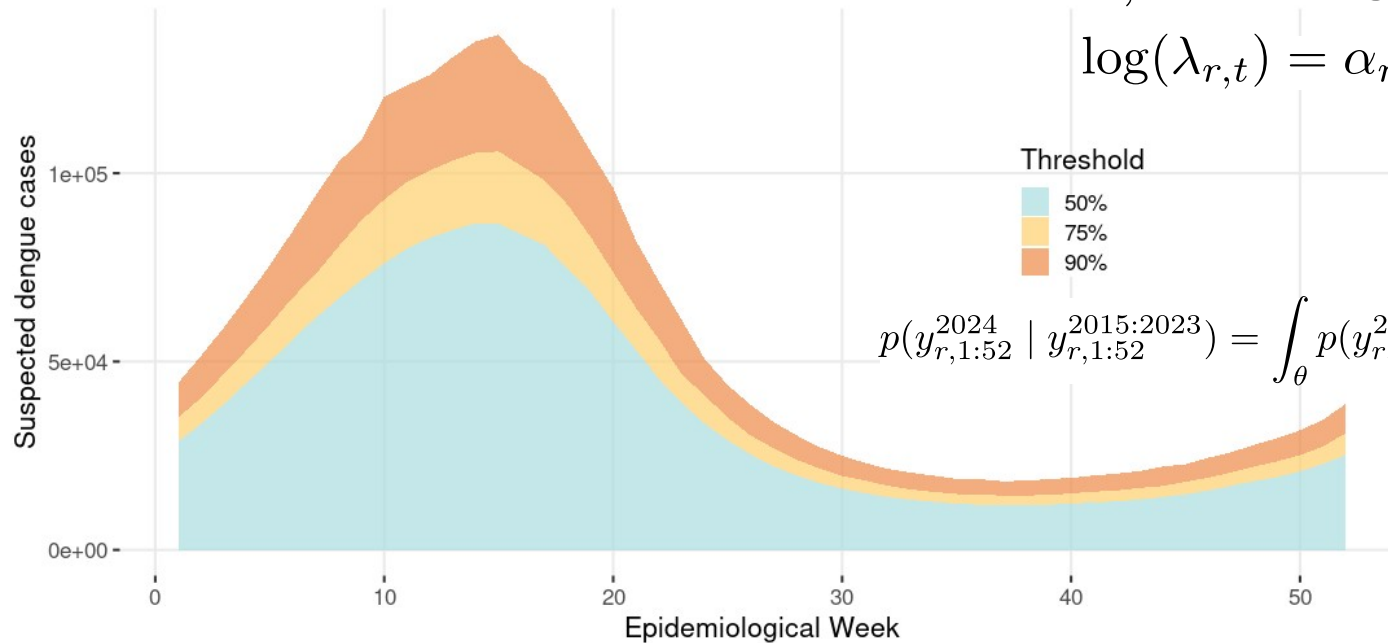
[Metrics](#)

[Preview PDF](#)

Forecasting probabilistic epidemic bands for dengue

$$Y_{r,t} \sim \text{NegBin}(\lambda_{r,t}, \phi_r)$$

$$\log(\lambda_{r,t}) = \alpha_r + \beta_{r,W[t]} + \gamma_{r,Y[t]}$$



$r=1,\dots,108$ macro de saude ;
 $t=1,2,3,\dots$; $W=1,2,\dots,52$;
 $Y=2015,2016,\dots$

$$p(y_{r,1:52}^{2024} | y_{r,1:52}^{2015:2023}) = \int_{\theta} p(y_{r,1:52}^{2024} | \theta) p(\theta | y_{r,1:52}^{2015:2023}) d\theta$$



Freitas et al. (2025)

Probabilistic epidemic bands

Table 1

Proposed classification of predicted cases level with the interpretation.

Band	Predicted cases level	Interpretation
$\leq 50\%$	Below the median, typical	Number of cases below the historical median, with a 50% occurrence probability.
50-75%	Moderately high, fairly typical	Number of cases moderately higher than the historical median, with a 25% occurrence probability.
75-90%	Fairly high, atypical	Number of cases fairly higher than the historical median, with a 15% occurrence probability.
$>90\%$	Exceptionally high, very atypical	Number of cases exceptionally higher than the historical median, with a 10% occurrence probability.

Freitas et al. (2025)

Training and predicting the season ahead

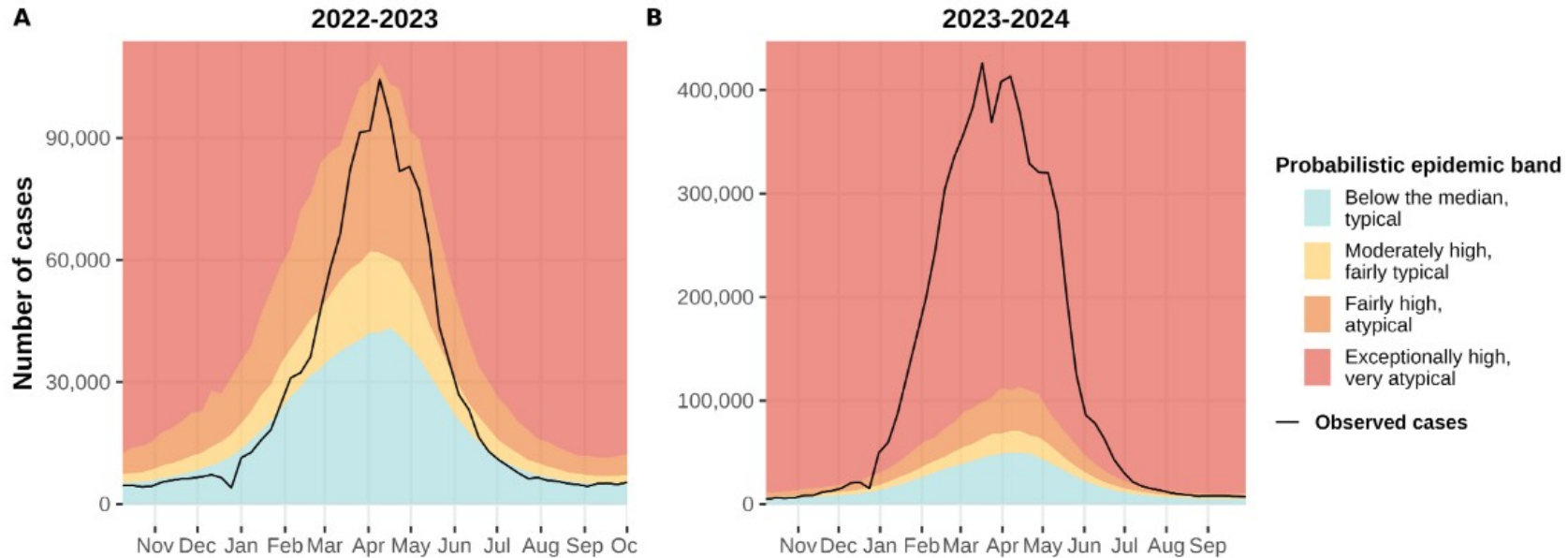


Figure 1: Estimated probabilistic epidemic bands compared with the observed number of dengue cases by week, 2022-2023 (A) and 2023-2024 (B) seasons, Brazil.

2024-2025

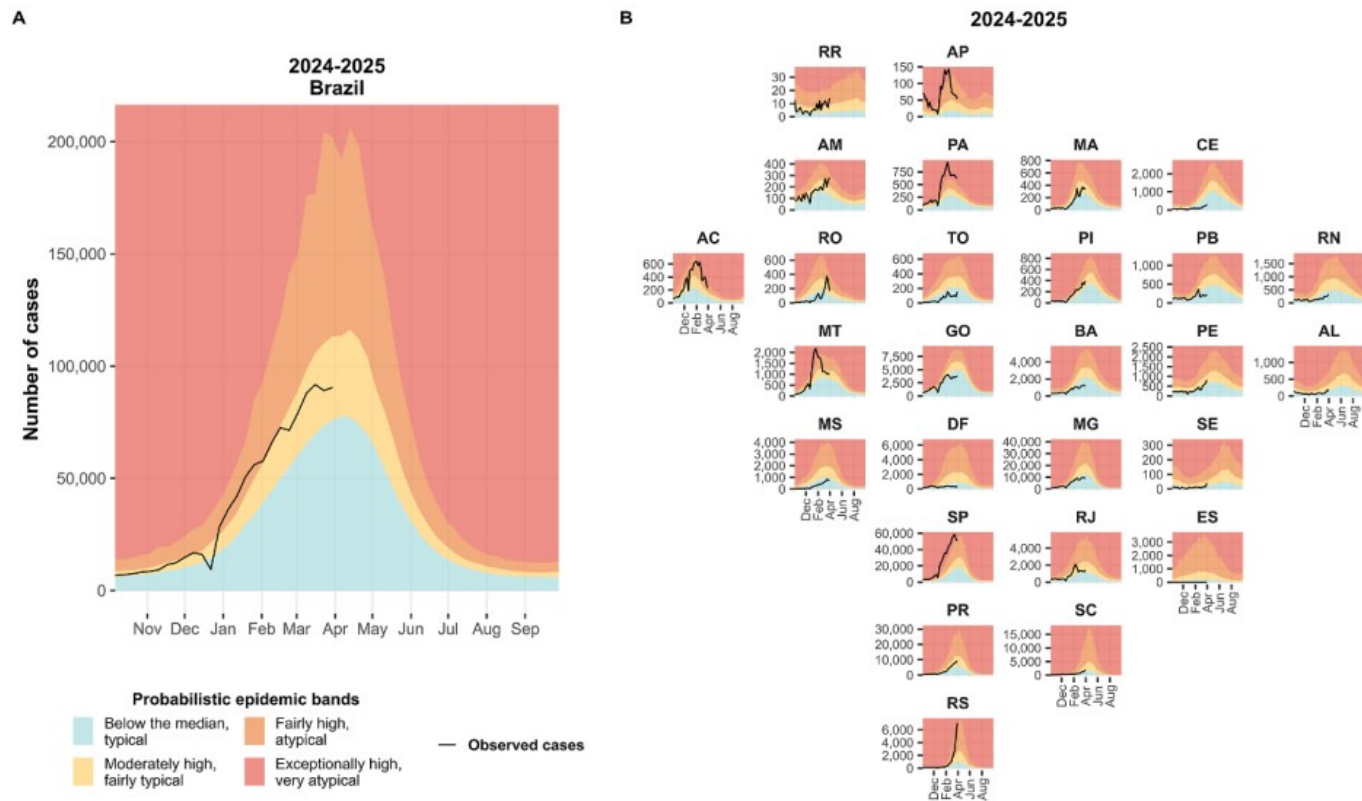
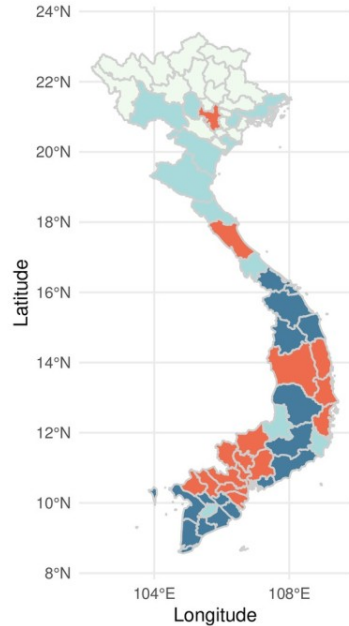


Fig. 4. Forecasted probabilistic epidemic bands of weekly dengue cases for the 2024–2025 season in Brazil (A) and by Federal Unit (B).

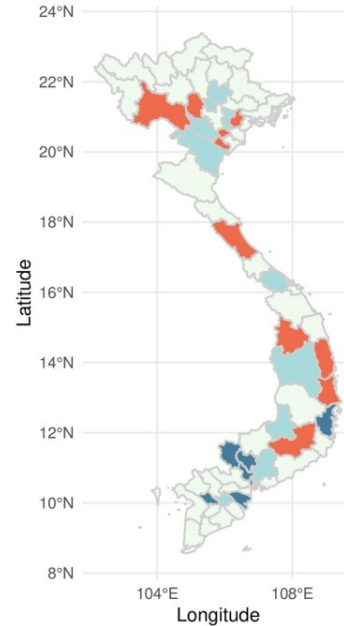
Dengue and clima: Predicting dengue in Vietnam

A



0-10 10-60 60-235 235-2490

B



0-0.2 0.2-0.4 0.6-0.8 0.8-1

Fig 8. Risk maps. (A) Spatial distribution of the posterior mean of the predicted number of dengue cases 1 month ahead for a forecast initialised on May 10, 2020. (B) Spatial distribution of the probability of exceeding an outbreak threshold based on the 95th percentile. The shapefile used to create this figure was obtained from DIVA-gis (<https://www.diva-gis.org>).

PLOS MEDICINE

RESEARCH ARTICLE

Probabilistic seasonal dengue forecasting in Vietnam: A modelling study using superensembles

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Colon-Gonzalez et al. (2020)



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Challenges and oportunities

- **Climate information**
 - Climate change (global warming)
 - Climate events (El Niño, La Niña)
 - Forecast weather and propagate uncertainty!
- **Immunity and control measures**
 - Previous epidemics
 - Vaccination
 - Wolbacchia
- **Other diseases and virus?**
 - COVID-19, RSV, Influenza, measles
- **Lagged variables**
 - DLNM
- **Dynamic models**
 - DQM for thresholds?
 - Spatially structured models
- **Statistical learning**
 - Forecasting and/or explain
 - Fast computation



Obrigado!



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